

DAN KLUSKIEWICZ

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Summary

Applied researcher with a geophysics background, building Python packages and statistical models for large remote-sensing problems, including individual-tree forest inventory from imagery, LiDAR, and field data.

Skills

Software: Python package development; scikit-learn, NumPy, pandas, GeoPandas, rasterio, laspy, Shapely

Geospatial data: aerial imagery, LiDAR point clouds, rasters, field observations, spatial sampling

Modeling: individual-tree segmentation, classification, and regression, with uncertainty estimates

Work Experience

Foresite — Moscow, ID (working in Index, WA)

September 2018 – Present

Programmer Statistician

Lead research and development for tools that predict individual-tree forest inventory from field data, aerial imagery, and LiDAR.

- Developed a reproducible end-to-end workflow that organizes multi-terabyte remote-sensing data, incorporates field data to segment individual trees and train models that classify them and predict tree size, evaluates those models with validation statistics, and scales inference across large areas.
- Built the Python package that runs this workflow and co-developed the geospatial library it uses for LiDAR, rasters, and tiling, including estimation and documentation of inventory uncertainty.
- Worked with researchers, engineers, foresters, and biometricians to maintain this workflow and to design field-sampling protocols that balance cost, logistics, and statistical precision.
- Presented inventory products and validation results to clients and at conferences (domestic and international).

Talus Analytics — Boulder, CO

June 2016 – April 2018

Quantitative Researcher

- Proposed a geometric method for probabilistic flood-hazard assessment and led its implementation in Python.
 - Implemented a simplified static wildfire-spread model from scratch, then extended it to time-dependent predictions.
 - Developed a wildfire accumulation model that delineates areas by hazard and predicts probable maximum losses within them.
 - Wrote the methods section for a FEMA publication on rapid flood modeling.
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Education

University of Washington

M.S. in Geophysics

2015

GPA: 3.94

NSF Graduate Research Fellowship

Pennsylvania State University, Schreyer Honors College

B.S. in Physics and Mathematics

2010

GPA: 3.99

Highest Distinction, Class rank **1** out of **519** (College of Science)

Publications

- Kluskiewicz, Dan and others. (2017). Sonic methods for measuring crystal orientation fabric in ice, and results from the West Antarctic ice sheet (WAIS) Divide. *Journal of Glaciology*. 63. 1–15. [doi:10.1017/jog.2017.20](https://doi.org/10.1017/jog.2017.20).